BIOL 2301: Genetics and Molecular Biology

Not an official Northeastern University syllabus. Course code, title and credits come from the institution's public catalog; everything else below is invented.

Credits	5 semester credits	**Term**	Fall 2024
Contact hours	75 hours (lecture + lab)	**Instructor**	Synthetic Instructor 6
Prerequisites	One introductory course in the discipline	**Office hours**	(not applicable - test file)

Course description

Genetics and Molecular Biology. Topics include organogenesis, gene expression, sensory systems, genomics and DNA replication; later weeks address mutation, genetic mapping and Mendelian genetics.

Learning outcomes

By the end of the course, students will be able to:
1. Explain the core ideas of organogenesis.
2. Apply gene expression to solve standard problems.
3. Analyze problems involving sensory systems.
4. Use genomics in practical exercises.
5. Evaluate approaches to DNA replication.
6. Communicate findings about mutation.

Weekly schedule

Week	Topic	Activity / assessment
1	Organogenesis	Problem set
2	Organogenesis (continued)	Lab or workshop
3	Gene expression	Quiz
4	Sensory systems	Case study
5	Sensory systems (continued)	Problem set
6	Genomics	Reading and discussion
7	Review and midterm	Midterm exam
8	DNA replication	Lab or workshop
9	DNA replication (continued)	Quiz
10	Mutation	Case study
11	Genetic mapping	Problem set
12	Genetic mapping (continued)	Reading and discussion
13	Mendelian genetics	Problem set
14	Review and synthesis	Final exam / project

Assessment

Component	Weight
Problem sets / assignments	25%
Labs	15%
Midterm exam	25%
Final exam or project	30%
Participation	5%

Materials and policies

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Required materials: instructor-provided open course notes and readings posted on the course site. Grading scale: A 93-100, A- 90-92, B+ 87-89, B 83-86, B- 80-82, C+ 77-79, C 73-76, C- 70-72, D 60-69, F below 60. Policies: see the institution's own catalog; none are reproduced here.

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